

TidyTuesday Week 39

Pranay Gundam

Thursday 24th September, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the little corner of the internet where we take two economic time series that have never met, force them into a spreadsheet together, and see if they hit it off. As always, the house rule stands: nothing here means anything, spurious correlation is the whole point, and if you use any of this to make a decision you deserve the outcome you get. Let's tour the wreckage.

Monday gave us New England commercial electricity producer prices versus a medical price index for, and I quote, "symptoms, signs, and ill-defined conditions." The raw regression came out swinging with an R-squared of 0.93, which is the kind of number that makes you briefly believe you've discovered something. Then we detrended it and the relationship evaporated so completely that the R-squared dropped to a heartwarming 0.02 with a p-value of basically one. In other words, both things went up over the years, and that's it. A textbook trend mirage. The medical category is literally named "ill-defined conditions," and honestly that's the most self-aware series we've ever regressed.

Tuesday paired active home listings in Flathead County, Montana with U.S. imports of maintenance and repair services. The raw regression was weak-ish (R-squared 0.12), which normally means we all go home disappointed. But plot twist: the detrended version got *stronger*, climbing to 0.19 and staying significant. This is the rare, genuinely-worth-a-raised-eyebrow case where the wiggles line up even after the shared drift is stripped out. Do I have a theory? Sure: absolutely none. Montana real estate agents fixing broken things they import? I refuse to explain it, and neither should you.

Wednesday's contestants were wholesale trade payrolls and the Dover, Delaware house price index, and this one actually holds up in both rounds – raw R-squared of 0.73, detrended R-squared of 0.73, both wildly significant. It's practically the same number twice, which for this project is downright suspicious in a good way. Payrolls and house prices both responding to the general state of the economy is at least a story a grown adult could tell with a straight face, so I'll allow it. Thursday, meanwhile, closed us out on a properly ridiculous note: Southwest-region central-bank GDP versus a discontinued Pacific-division bank-assets ratio. The raw result limped in barely-not-significant (p around 0.055, the statistical equivalent of "we need to talk"), but detrending flipped it into significance at 0.008. Both regressions came with a stern warning about numerical problems and enormous condition numbers, which is the software's polite way of saying "are you sure about this." I am not. I never am. See you next week for more of the same.

2 Date: 2026-09-21

Series ID: PCU221122221122421

This series is titled Producer Price Index by Industry: Electric Power Distribution: Commercial Electric Power for New England Census Division and has a frequency of Monthly. The units are Index Dec 1990=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1971-01-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

Series ID: SYSIILPIBLEND

This series is titled Medical Services Expenditures by Disease: Symptoms; Signs; and Ill-Defined Conditions Price Index, Blended Account Basis and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-01-01 and the observation end date is 2021-01-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_SYSIILPIBLEND	R-squared:	0.933
Model:	OLS	Adj. R-squared:	0.929
Method:	Least Squares	F-statistic:	277.7
Date:	Mon, 21 Sep 2026	Prob (F-statistic):	3.41e-13
Time:	17:16:08	Log-Likelihood:	-68.300
No. Observations:	22	AIC:	140.6
Df Residuals:	20	BIC:	142.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-6.0876	5.281	-1.153	0.263	-17.103	4.928
value_fred_PCU221122221122421	0.5458	0.033	16.664	0.000	0.477	0.614

Omnibus:	6.881	Durbin-Watson:	1.268
Prob(Omnibus):	0.032	Jarque-Bera (JB):	5.911
Skew:	0.534	Prob(JB):	0.0521
Kurtosis:	5.304	Cond. No.	706.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

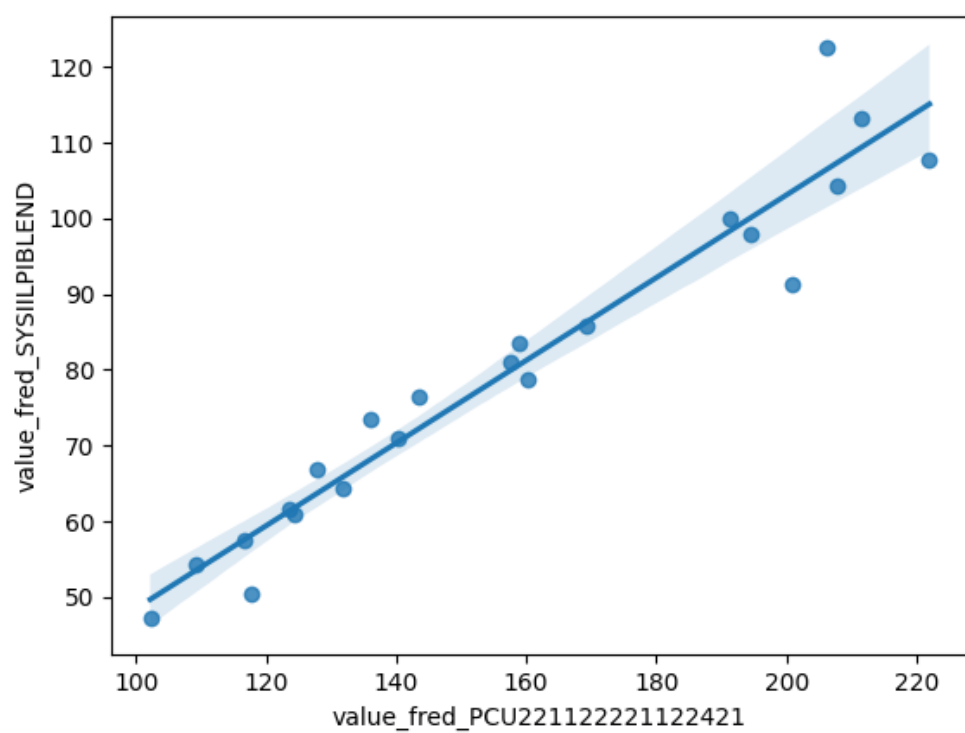


Figure 1: Raw Regression Plot for 2026-09-21

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_SYSIILPIBLEND_detrended	R-squared:	0.022
Model:	OLS	Adj. R-squared:	-0.027
Method:	Least Squares	F-statistic:	0.4428
Date:	Mon, 21 Sep 2026	Prob (F-statistic):	0.513
Time:	17:16:08	Log-Likelihood:	-54.441
No. Observations:	22	AIC:	112.9
Df Residuals:	20	BIC:	115.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.119e-14	0.643	6.41e-14	1.000	-1.340	1.340
value_fred_PCU221122221122421_detrended	0.0480	0.072	0.665	0.513	-0.103	0.199

Omnibus:	12.738	Durbin-Watson:	0.457
Prob(Omnibus):	0.002	Jarque-Bera (JB):	12.678
Skew:	1.200	Prob(JB):	0.00177
Kurtosis:	5.841	Cond. No.	8.90

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

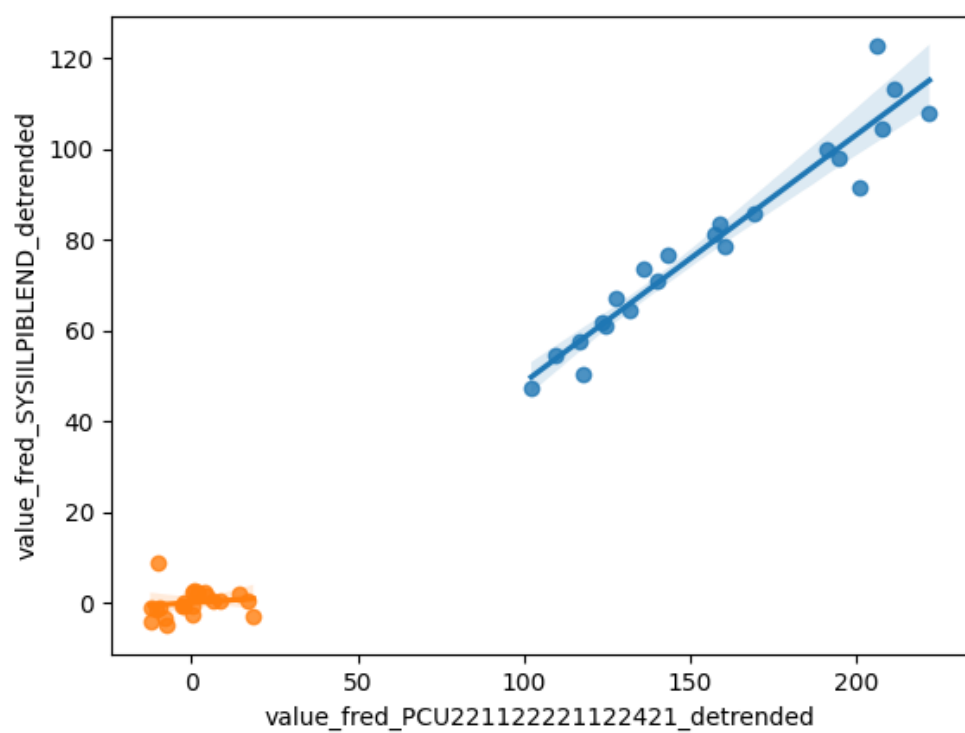


Figure 2: Detrended Regression Plot for 2026-09-21

3 Date: 2026-09-22

Series ID: ACTLISCOU30029

This series is titled Housing Inventory: Active Listing Count in Flathead County, MT and has a frequency of Monthly. The units are Level and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2016-07-01 and the observation end date is 2026-08-01. The popularity of this series is 2.

Series ID: ITMMARM133S

This series is titled U.S. Imports of Services: Maintenance and Repair Services, not included elsewhere and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1999-01-01 and the observation end date is 2026-07-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ITMMARM133S	R-squared:	0.123
Model:	OLS	Adj. R-squared:	0.116
Method:	Least Squares	F-statistic:	16.76
Date:	Tue, 22 Sep 2026	Prob (F-statistic):	7.76e-05
Time:	15:49:46	Log-Likelihood:	-747.35
No. Observations:	121	AIC:	1499.
Df Residuals:	119	BIC:	1504.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	452.9830	36.286	12.484	0.000	381.134	524.832
value_fred_ACTLISCOU30029	0.1837	0.045	4.094	0.000	0.095	0.273

Omnibus:	12.594	Durbin-Watson:	0.214
Prob(Omnibus):	0.002	Jarque-Bera (JB):	14.022
Skew:	0.833	Prob(JB):	0.000902
Kurtosis:	3.056	Cond. No.	2.75e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.75e+03. This might indicate that there are strong multicollinearity or other numerical problems.

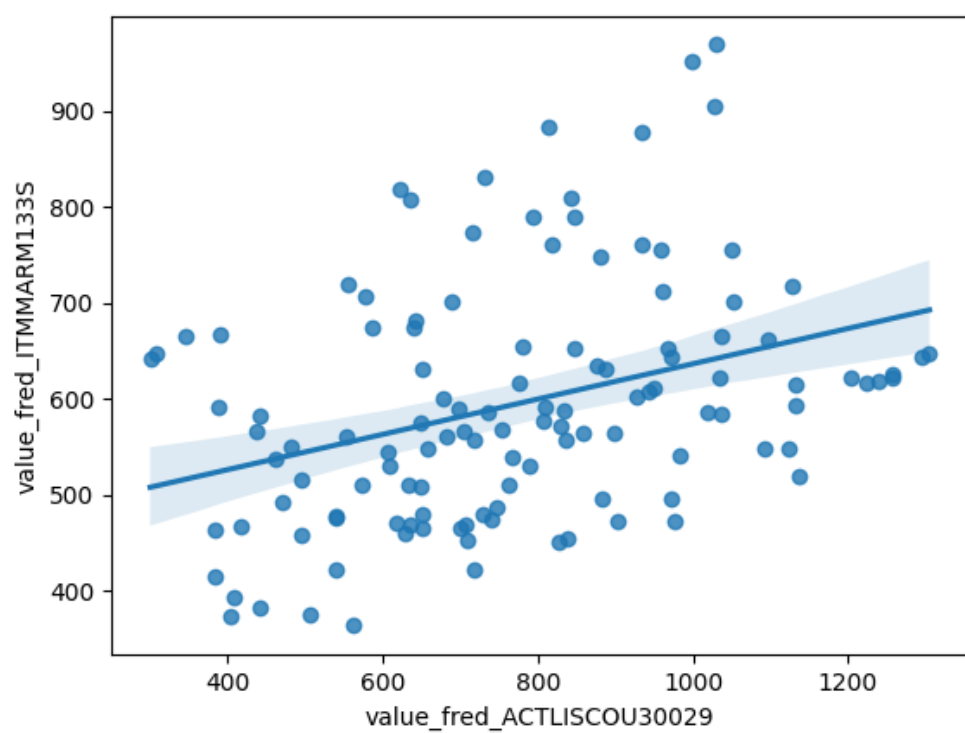


Figure 3: Raw Regression Plot for 2026-09-22

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ITMMARM133S_detrended	R-squared:	0.195
Model:	OLS	Adj. R-squared:	0.188
Method:	Least Squares	F-statistic:	28.77
Date:	Tue, 22 Sep 2026	Prob (F-statistic):	4.07e-07
Time:	15:49:46	Log-Likelihood:	-740.70
No. Observations:	121	AIC:	1485.
Df Residuals:	119	BIC:	1491.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-3.037e-11	10.104	-3.01e-12	1.000	-20.008	20.008
value_fred_ACTLISCOU30029_detrended	0.2437	0.045	5.364	0.000	0.154	0.334

Omnibus:	4.932	Durbin-Watson:	0.254
Prob(Omnibus):	0.085	Jarque-Bera (JB):	5.006
Skew:	0.473	Prob(JB):	0.0819
Kurtosis:	2.689	Cond. No.	222.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

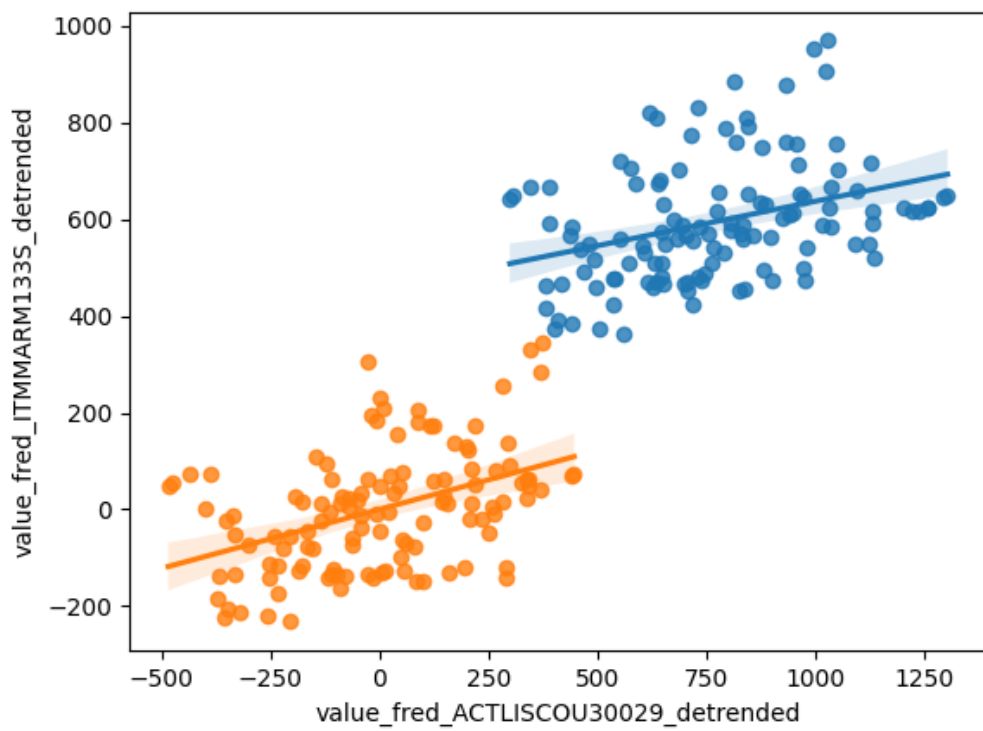


Figure 4: Detrended Regression Plot for 2026-09-22

4 Date: 2026-09-23

Series ID: CEU4142000017

This series is titled Indexes of Aggregate Weekly Payrolls of All Employees, Wholesale Trade and has a frequency of Monthly. The units are Index 2007=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2006-03-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

Series ID: ATNHPIUS20100Q

This series is titled All-Transactions House Price Index for Dover, DE (MSA) and has a frequency of Quarterly. The units are Index 1995:Q1=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1987-07-01 and the observation end date is 2026-04-01. The popularity of this series is 2.

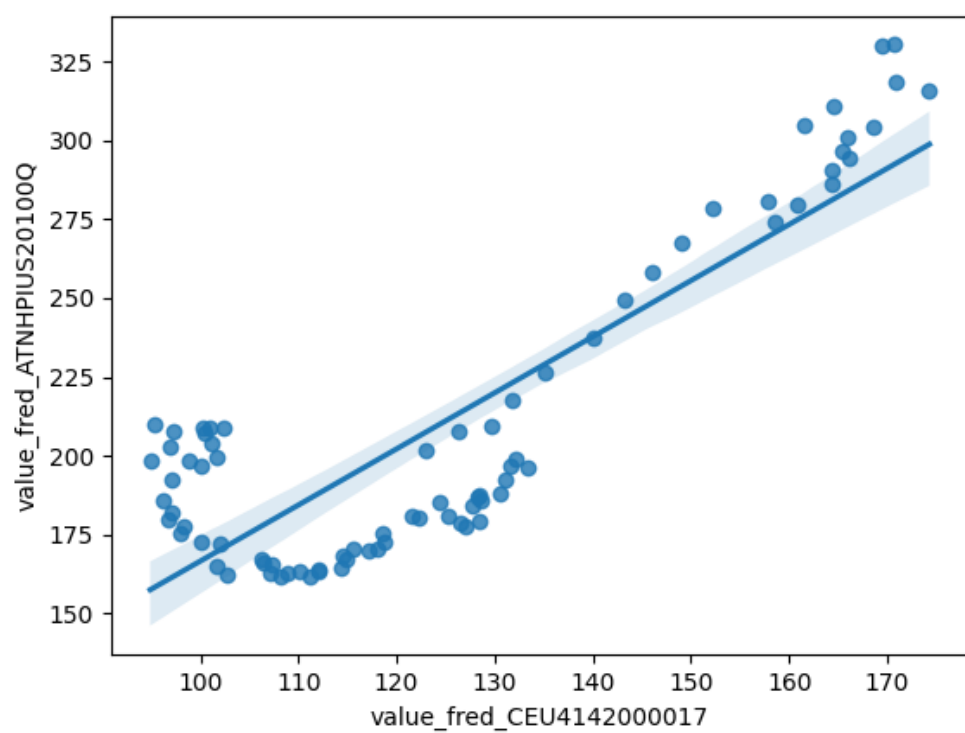
4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ATNHPIUS20100Q	R-squared:	0.731			
Model:	OLS	Adj. R-squared:	0.728			
Method:	Least Squares	F-statistic:	214.8			
Date:	Wed, 23 Sep 2026	Prob (F-statistic):	3.06e-24			
Time:	15:39:04	Log-Likelihood:	-377.92			
No. Observations:	81	AIC:	759.8			
Df Residuals:	79	BIC:	764.6			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-11.5081	15.435	-0.746	0.458	-42.230	19.214
value_fred_CEU4142000017	1.7805	0.121	14.656	0.000	1.539	2.022
Omnibus:	32.589	Durbin-Watson:	0.083			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	6.325			
Skew:	0.272	Prob(JB):	0.0423			
Kurtosis:	1.744	Cond. No.	678.			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ATNHPIUS20100Q_detrended	R-squared:	0.727
Model:	OLS	Adj. R-squared:	0.724
Method:	Least Squares	F-statistic:	210.7
Date:	Wed, 23 Sep 2026	Prob (F-statistic):	5.32e-24
Time:	15:39:04	Log-Likelihood:	-349.22
No. Observations:	81	AIC:	702.4
Df Residuals:	79	BIC:	707.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-7.785e-12	2.029	-3.84e-12	1.000	-4.039	4.039
value_fred_CEU4142000017_detrended	4.4163	0.304	14.517	0.000	3.811	5.022

Omnibus:	7.128	Durbin-Watson:	0.420
Prob(Omnibus):	0.028	Jarque-Bera (JB):	3.489
Skew:	0.261	Prob(JB):	0.175
Kurtosis:	2.128	Cond. No.	6.67

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

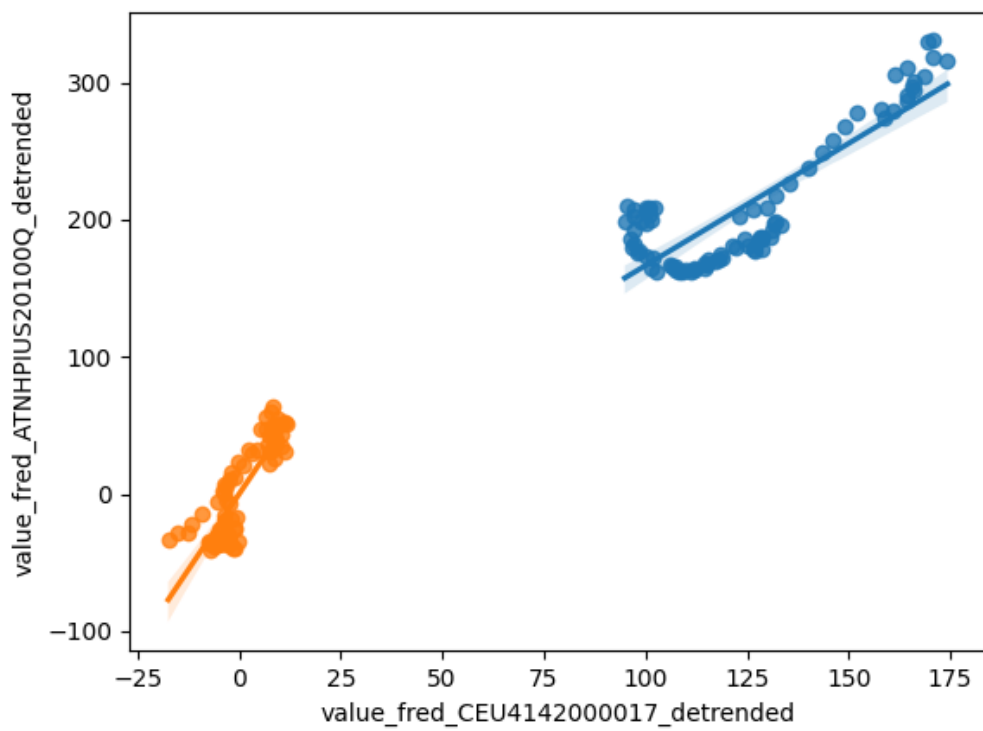


Figure 6: Detrended Regression Plot for 2026-09-23

5 Date: 2026-09-24

Series ID: SWSTFRBCIRGSP

This series is titled Real Gross Domestic Product: Monetary Authorities-Central Bank, Credit Intermediation, and Related Services (521-522) in the Southwest BEA Region and has a frequency of Annual. The units are Millions of Chained 2017 Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1997-01-01 and the observation end date is 2024-01-01. The popularity of this series is 1.

Series ID: LLRNPT39

This series is titled Assets at Banks whose ALLL exceeds their Nonperforming Loans, Banks with Total Assets from \$1B to \$10B, Pacific Census Division (DISCONTINUED) and has a frequency of Quarterly, End of Period. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1988-01-01 and the observation end date is 2020-07-01. The popularity of this series is 1.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_LLRNPT39	R-squared:	0.158
Model:	OLS	Adj. R-squared:	0.119
Method:	Least Squares	F-statistic:	4.120
Date:	Thu, 24 Sep 2026	Prob (F-statistic):	0.0546
Time:	15:59:56	Log-Likelihood:	-110.56
No. Observations:	24	AIC:	225.1
Df Residuals:	22	BIC:	227.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	152.7995	40.101	3.810	0.001	69.635	235.964
value_fred_SWSTFRBCIRGSP	-0.0016	0.001	-2.030	0.055	-0.003	3.37e-05

Omnibus:	4.915	Durbin-Watson:	0.325
Prob(Omnibus):	0.086	Jarque-Bera (JB):	3.660
Skew:	-0.955	Prob(JB):	0.160
Kurtosis:	3.111	Cond. No.	4.07e+05

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

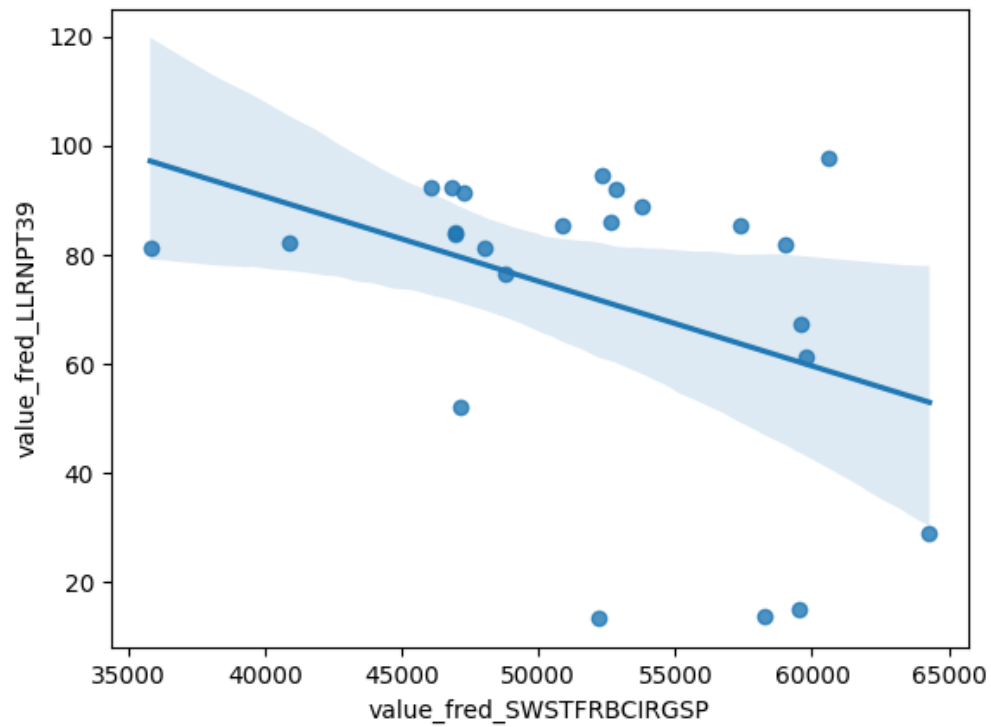


Figure 7: Raw Regression Plot for 2026-09-24

[2] The condition number is large, $4.07e+05$. This might indicate that there are strong multicollinearity or other numerical problems.

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_LLRNPT39_detrended	R-squared:	0.277
Model:	OLS	Adj. R-squared:	0.244
Method:	Least Squares	F-statistic:	8.412
Date:	Thu, 24 Sep 2026	Prob (F-statistic):	0.00830
Time:	15:59:56	Log-Likelihood:	-108.67
No. Observations:	24	AIC:	221.3
Df Residuals:	22	BIC:	223.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.199e-12	4.776	2.51e-13	1.000	-9.904	9.904
value_fred_SWSTFRBCIRGSP_detrended	-0.0031	0.001	-2.900	0.008	-0.005	-0.001

Omnibus:	7.727	Durbin-Watson:	0.515
Prob(Omnibus):	0.021	Jarque-Bera (JB):	5.892
Skew:	-1.175	Prob(JB):	0.0526
Kurtosis:	3.608	Cond. No.	4.43e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 4.43e+03. This might indicate that there are strong multicollinearity or other numerical problems.

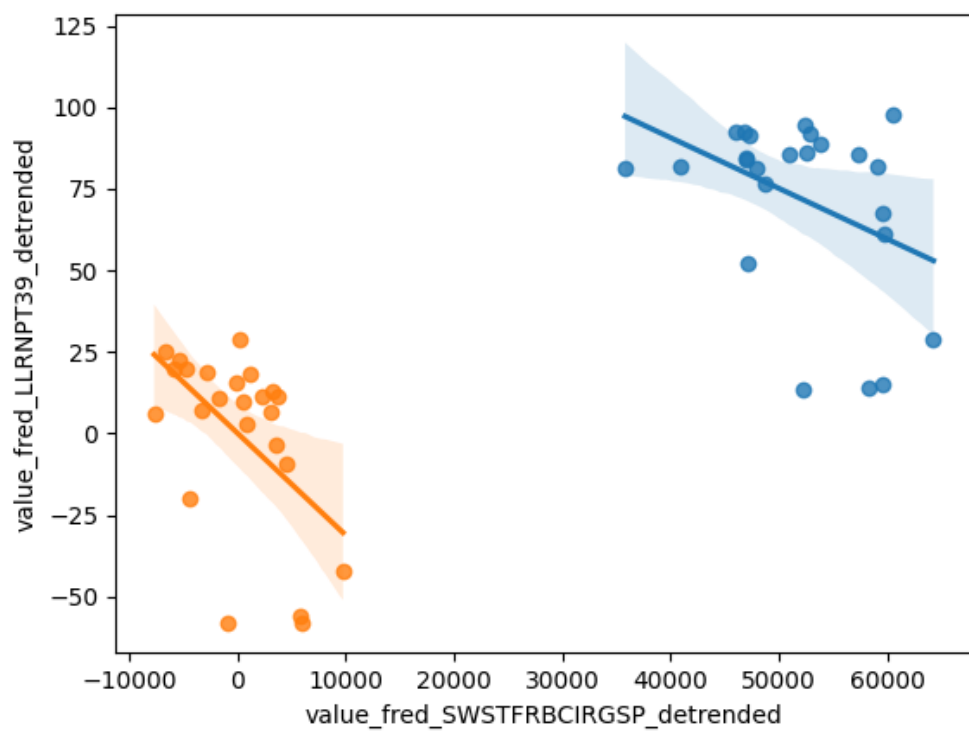


Figure 8: Detrended Regression Plot for 2026-09-24